

Single-Photon Double Slit: Experiment III

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Physics 5700

22 Apr 2025

Abstract—Thomas Young’s famous double slit experiment demonstrated the wave nature of light (1801). The discovery of the photoelectric effect (Hertz, 1887) demonstrated the particle nature of light. These seemingly opposing perspectives introduced the need for a new construct in physics: *wave-particle duality*. Here, we recreate the original experiment with an apparatus designed to perform quantitative analysis of *Frauenhofer* diffraction patterns using two main setups: (1) with a laser and a photodetector, and (2) with a low-intensity bulb and a photomultiplier tube. The experiment allows us to demonstrate that (1) the apparatus’ slit width and slit separation are $a = 0.071 \pm 0.001$ mm and $b = 0.410 \pm 0.005$ mm, respectively, and (2) *Frauenhofer* diffraction occurs with both setups, despite the high unlikelihood that more than a single photon is present in the apparatus at one time, which begs an important question of quantum mechanics: what is the photon interfering with?

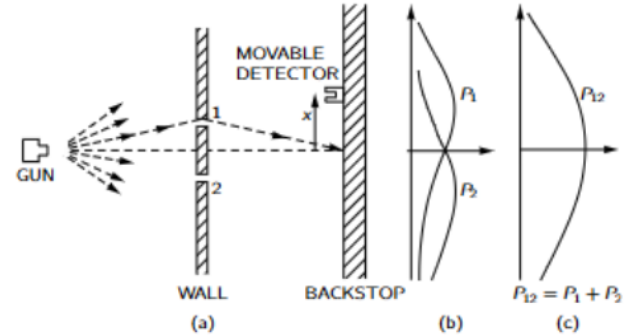


Fig. 1. See Ref. [2]. Expected spatial distribution of classical particles propagating through a double slit apparatus.

I. INTRODUCTION

The original double slit experiment performed by Thomas Young (1801) is one of the most well known experiments in the realm of optics and quantum mechanics, as it was one of the first experiments which demonstrated the *wave-like* nature of light.

The demonstration of *interference patterns* by Young contested the previously hypothesized “corpuscles” of light. Especially after the discovery of the photoelectric effect, which demonstrated *particle-like* behavior, the *wave-particle duality* of light was disputed for many years. This argument was fundamental to the development of quantum mechanics.

In particular, this experiment aims to demonstrate both wave-like and particle-like phenomena of light by performing the double slit experiment in such low intensity conditions that individual photons are present in the apparatus at a moment in time.

Expectations are such that individual *impulsive signals* can be detected and viewed with an oscilloscope such that a single “packet” of light—a photon—can be observed at one time. Each event demonstrates particle behavior. The spatial distribution of the locations of photon arrivals at a detector should reveal an interference pattern which is suggestive of wave behavior. The presence of only a single photon in the apparatus at one time can be essentially guaranteed by a statistical argument based on the rate of photon arrivals. This prompts the question of how such interference occurs even when only one photon is present (See Ref. [1]).

II. THEORY

Consider a gun, which fires “classical” particles through an apparatus containing two slits. If the slits are wide enough to allow particles through, and the “wall” is strong enough

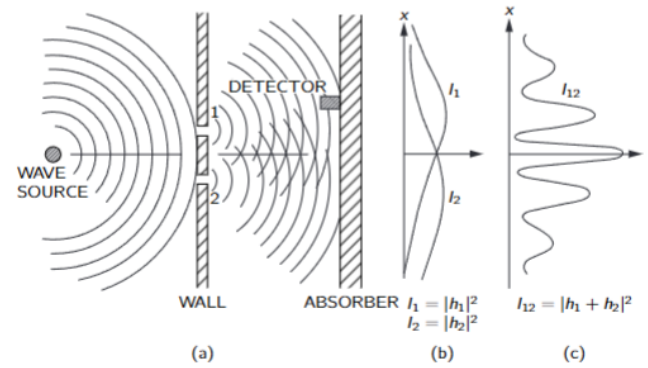


Fig. 2. See Ref. [2]. Illustration of how wave-like sources produce a diffraction pattern with constructive and destructive interference.

to stop particles which don’t fall directly upon the slits, one might expect to see a probability distribution like those on the right side of figure 1 (See Ref. [2]).

Because light waves interfere with each other, two separate sources will produce a diffraction pattern such that there are observable “bright” and “dark” regions. These regions occur where the waves from each source add *constructively* and *destructively*.

The predictions of the *Frauenhofer* diffraction theory produce a function of diffraction pattern intensity with respect to angular position, $I(\theta)$. For *single slit diffraction*,

$$I(\theta) = I_0 \cdot \left(\frac{\sin \alpha}{\alpha} \right)^2 \quad (1)$$

and in the case of *double slit diffraction*,

$$I(\theta) = 4I_0 \cdot \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot \cos^2 \beta \quad (2)$$

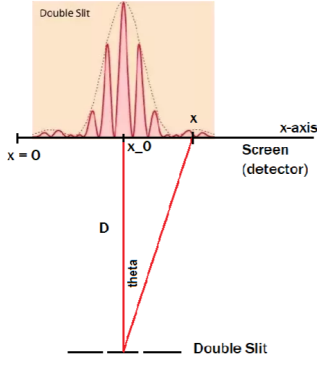


Fig. 3. Schematic diagram labeling angles and distances relating to the positions of the double slit and detector or screen, as well as a quantitative image illustrating the expected interference pattern. (See Ref. [3]).

where $\alpha = \frac{ka}{2} \sin \theta$ and $\beta = \frac{kb}{2} \cos \theta$. The slit separation is b , while the slit width is a . The *propagation number* is defined as $k = 2\pi/\lambda$. For small angles, we then have the following relationships:

$$\begin{aligned} \alpha &= \frac{\pi a}{\lambda D} \\ \beta &= \frac{\pi b}{\lambda D} \end{aligned} \quad (3)$$

At a specified distance, D , from the double slit, the spatial position, x , of the detector is related to the angle, θ , by

$$\tan \theta = \frac{x - x_0}{D} \approx \sin \theta \quad (4)$$

for small angles. See Ref. [4] for derivations.

Eventually, it will become evident that the Fraunhofer diffraction pattern appears over any significant enough time duration in the case of the low-intensity light bulb which emits individual photons at a time. We can make a statistical argument based on the rate of photon arrivals, Γ , the efficiency of the PMT detector, η , and the length of the apparatus, L .

A single photon has a time of flight, $\tau = L/c$. For certain efficiency, η , we can calculate the true rate of photon events via

$$\Gamma = \frac{\gamma}{\eta} \quad (5)$$

where Γ is the true number of photons which propagate through the apparatus per unit time, γ is the number of photon arrival events actually detected per unit time, and η is the fraction of photons which the PMT actually detects.

The probability of a single photon being present in the apparatus at any given time is then given by

$$\mathcal{P}(1) = \frac{\tau}{\Gamma} \quad (6)$$

Since two photon events are independent of each other, the probability that two photons are present in the apparatus at once is given by the square of this value.

$$\mathcal{P}(2) = \left(\frac{\tau}{\Gamma} \right)^2 \quad (7)$$

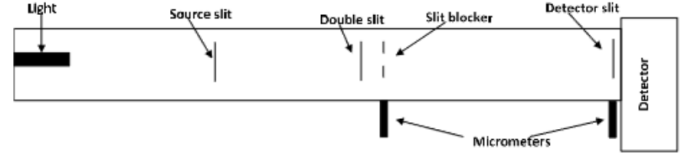


Fig. 4. Schematic diagram of the double-slit apparatus. The light source may either be a 5 mW laser or a low intensity lamp. The detector may be either a photodetector or a photomultiplier tube (the PMT is compatible only with the low-intensity bulb). A source slit is used to homogenize the light source, which is especially important when the lamp is in use.

III. METHODS

The apparatus consists of a light source, from which light propagates through a source slit in order to homogenize the source and ensure that plane waves are propagating through the instrument. From the source slit, light then propagates through a double slit, behind which a slit blocker is used to block either slit at once. Finally, the light falls on a detector slit used for selective measurement of the spatial distribution of light intensity. Behind the detector slit is either a *photodetector*, which produces a voltage that scales linearly with the intensity of the light absorbed, or a *photomultiplier tube* (PMT), which uses the photoelectric effect to generate significant voltage pulses from individual photon detection events.

Initially, some alignment is necessary in order to produce the diffraction pattern in the correct location such that it can be observed with the detector. Follow the steps highlighted in Ref. [5]. Additionally, we note the micrometer values of five separate locations of the slit blocker such that the slit blocker can be operated while the apparatus is *closed* (See Ref. [1]):

- 1) Both slits are blocked.
- 2) Light emerges from the farther of two slits.
- 3) Light emerges from both slits.
- 4) Light emerges from the nearer of two slits.
- 5) Both slits are blocked (highest reading).

The experiment is performed three different times, each with slight modifications. Initially, it is performed qualitatively with a laser (670 ± 5 nm)—an index card is placed at the detector end to visually observe the interference pattern while the instrument enclosure is *open*, allowing the experimenter to demonstrate the constructive and destructive addition of waves from both slits. Additionally, the slits which are open can be controlled to show that subtracting one light source from the system does in fact eliminate the fringe pattern. The pattern seen here is that of *Fraunhofer* diffraction.

The second portion of the experiment is a repetition of the first part, but with a more quantitative approach that aims to perform accurate measurements of the pattern intensity and its spatial distribution. Measurements are obtained by shifting the “detector slit” (which permits light to pass through to the photodetector) across the diffraction pattern. This allows for a spatial mapping of the intensity distribution which can be compared to theoretical predictions of *Fraunhofer diffraction*.

The third and final sub-experiment requires a bulb with extremely low intensity output. A filter is placed over the bulb such that photons are only produced in the small wavelength range of 541 to 551 nm.

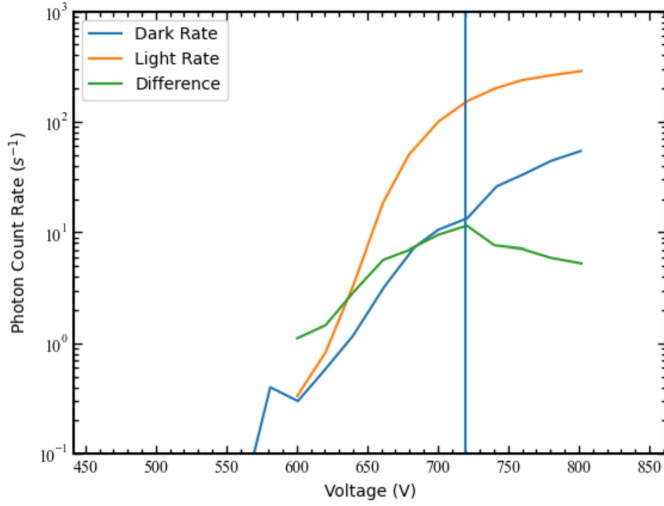


Fig. 5. Logarithmic plot of bright rate (central maximum, both slits unblocked) and dark rate (central maximum, both slits blocked) of photon events, plotted over a range of PMT voltage biases. The voltage where these curves are separated furthest is the ideal voltage for the PMT bias.

The PMT creates detectable voltage pulse signals from the arrival of individual photons which propagate through the diffracting apparatus. The amplitude of the analog voltage pulse is set by an amplification bias on the PMT, which helps to further isolate background noise from actual photon events. This analog pulse is then sent to a digitizer with a fixed voltage threshold which converts the analog pulse to a digital square wave pulse. The TTL counter then records the number of digitized photon events each second, which serves as the dependent variable in our measurements.

This sub-experiment will demonstrate that light does indeed arrive in finite, quantized packets as particles. Still, the observation of these photons demonstrates the Fraunhofer diffraction predictions as a spatial probability distribution of photon arrivals, despite only a single photon being present in the apparatus at one time. (See Ref. [1] and [5] for full instructions on taking measurements). This wave-particle duality, of course, is a fundamental question of quantum mechanics.

The voltage bias on the PMT was chosen by comparing a “bright” rate observed on the TTL counter compared to a “dark” rate of photon events. The voltage with the largest ratio of bright rate to dark rate (largest logarithmic separation as shown in figure 5) was chosen to ensure minimal background effects. As is evident in the figure, the ideal voltage bias was fixed at 720 V for the remainder of the experiment.

The reason for obtaining this ideal voltage bias is because we want to be sure that we are producing as many analog pulses as possible that reach the threshold of the digitizer—while minimizing “background” events that can also reach this threshold. This ensures that we are accurately counting true photon arrival events, without missing too many or including too much noise.

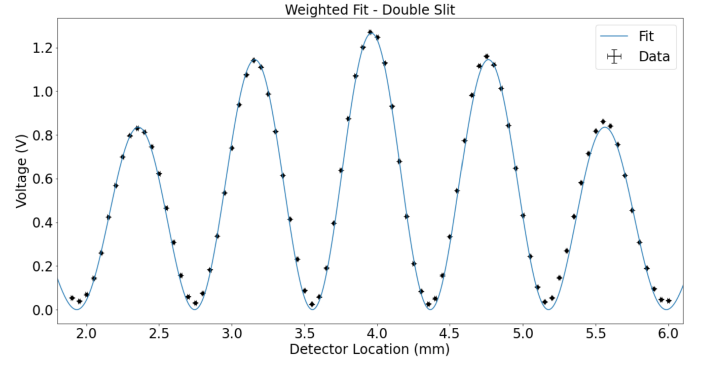


Fig. 6. Recorded data for position (3), with both slits allowing light through. The expected Fraunhofer diffraction pattern is evidently present, and the curve was fit with $\chi^2 = 1.56$

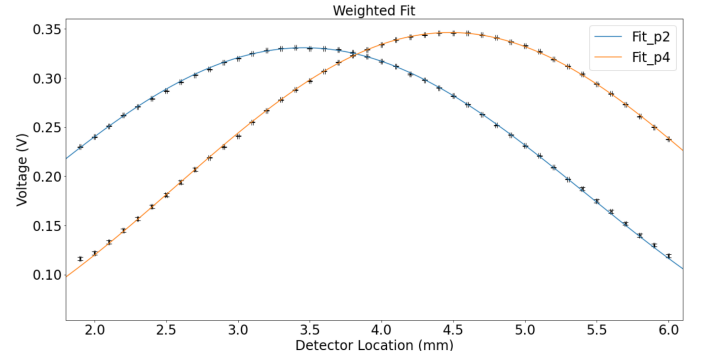


Fig. 7. Data and corresponding fits of equation (1) for single slit data from the left slit (blue) and right slit (orange). As expected, there is a difference between locations of the central peak due to the fact that the slits are separated horizontally. The resulting fit agrees very strongly with our measurements, producing $\chi^2_{\text{left}} = 0.0020$ and $\chi^2_{\text{right}} = 0.0031$.

IV. ANALYSIS & INTERPRETATION

The goal of analysis in the laser portion of the experiment is to demonstrate that the double slit apparatus does in fact produce diffraction patterns which coincide with the predictions of Fraunhofer diffraction theory, and then use those predictions to estimate the dimensional characteristics of the double slit, such as slit separation and slit width.

We can perform a fit to the experimental data obtained by measuring the photodetector’s output voltage plotted against the detector slit position (in millimeters). Fitting data at positions (2) and (4) to equation (1) produces two estimates for the fit parameter α . Fitting of the data at position (3) to equation (2) produces an estimate of fit parameter β and a third estimate of the fit parameter α . The data and resulting fits are shown in Figures 6 and 7.

Note that the amplitude of the left and right fit are in slight disagreement, which suggests a small alignment issue with the apparatus. This is expected to introduce some systematic error into the final results.

As expected, the classically predicted bright and dark fringes appear, as produced by constructive and destructive interference of waves from each slit.

The resulting estimate of fit parameters from the fits on

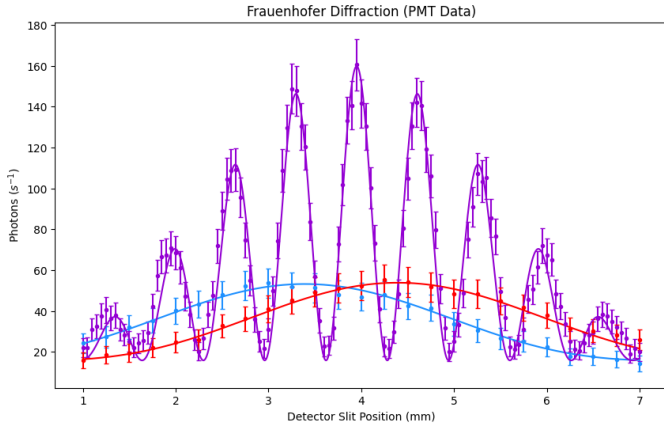


Fig. 8. PMT data plotted similarly to photodetector data for positions (2), (3), and (4) (blue, purple, red). Photon counts per second averaged over ten 1-second intervals, with error given as $\sqrt{N}$ as photon arrivals are independent events. Fit parameters obtained from laser data were used to generate the plotted curves, with the new wavelength accounted for. Even though only a single photon is present at a time in the apparatus, the Fraunhofer diffraction pattern is still clearly observable.

positions (2), (3), and (4) are as follows:

$$\begin{aligned}\alpha &= 0.678 \pm 0.005 \text{ mm}^{-1} \\ \beta &= 3.876 \pm 0.007 \text{ mm}^{-1}\end{aligned}\quad (8)$$

Using equation (3), we estimate the dimensionality of the double slit and propagate the error from uncertainties on the fit parameters α and β as well as λ and D .

$$\begin{aligned}a &= 0.071 \pm 0.001 \text{ mm} \\ b &= 0.410 \pm 0.005 \text{ mm}\end{aligned}\quad (9)$$

Unfortunately, the prediction of a differs from the provided result by quite a bit. In fact, we have underestimated here by about 14σ , as $\langle a \rangle = 0.085 \text{ mm}$. The large discrepancy is likely due to accumulated error from misalignment of the apparatus (as previously noted by the disagreement between single slit amplitudes). The slit separation parameter, however, is in strong agreement with one of three possible expectation values. We assume that the true value is the one closest to our estimate, $\langle b \rangle = 0.406 \text{ mm}$, thus the true value falls within 1σ of our estimate of slit separation.

The third part of the experiment requires a slightly different approach. Individual photons were counted with the PMT, oscilloscope, and TTL counter over ten 1-second intervals with a PMT bias of 720 V (see previous discussion and figure 4). Photons arrived at some low background rate in the “dark” areas of the fringe pattern, (which adds some constant vertical offset to the data) and at some much higher rate in the “bright” regions.

Effectively, the process of measurement is identical, but the results now provide the amplitude (dependent variable) in units of an event rate, s^{-1} , as opposed to the previous voltage generated by the photodetector.

As expected, the Fraunhofer diffraction pattern is present, and even agrees somewhat well with fit parameters produced by the laser experiment.

Seeing as the average rate of photon arrivals is on the order of 10^2 s^{-1} , the photon travel distance is on the order of 1 m, the efficiency of the PMT is around 4%, then the probability that more than one photon is present in the apparatus at any given moment is extremely insignificant:

$$\mathcal{P}(2) \approx 2 \times 10^{-24} \quad (10)$$

V. CONCLUSIONS

In this experiment we have used a photodetector voltage amplitude plotted against detector position to spatially map the Fraunhofer diffraction pattern in order to (1) demonstrate the wave-like nature of homogenized laser light ($670 \pm 5 \text{ nm}$) and (2) estimate the slit width and slit separation of a double slit apparatus.

$$\begin{aligned}a &= 0.710 \pm 0.001 \text{ mm} \\ b &= 0.410 \pm 0.005 \text{ mm}\end{aligned}\quad (11)$$

The expectation values of these predictions fall within 14σ and 1σ of our estimates, respectively, suggesting significant systematic errors were introduced in particular for estimation of the slit width parameter.

$$\begin{aligned}\langle a \rangle &= 0.085 \text{ mm} \\ \langle b \rangle &= 0.406 \text{ mm}\end{aligned}\quad (12)$$

This systematic error is evidenced in the difference between peak probability amplitudes for the pattern produced at slit blocker positions (2), with the right slit blocked and (4), with the left slit blocked, implying that more light was able to propagate through one slit than the other.

We have also demonstrated the particle-like nature of light from an extremely low intensity bulb, which can produce individual photons in the apparatus. These photon arrival events are observed with the photoelectric effect—a well known example of particle-like behavior—and are then counted over a fixed time interval to obtain a rate of arrivals as well as another spatially mapped Fraunhofer diffraction pattern—a well known example of wave-like behavior. This *duality* is entirely foreign to classical human perception, and it prompts a classically unanswerable question: how can something be a *particle* and a *wave* at the same time?

Based on the rate of photon arrivals, we argued that the probability of more than a single photon being present in the apparatus at any one moment was extremely unlikely ($\mathcal{P} \approx 10^{-24}$). However, Fraunhofer diffraction implies that each individual photon which propagates through the apparatus must be interfering with *something*, despite the statistical unlikelihood (near impossibility) that any other photon is present. This result of the experiment only poses another paradoxical question to be answered by quantum mechanics.

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- [2] R. Feynman, *The Feynman Lectures on Physics, Volume III*. Pearson, 1964.
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- [5] “2-Slit Interference, 1 Photon at a Time v2.3,” *CarmenCanvas*, 2022.